

Polynomial Arithmetic

Answer Key

Algebra 2

Quick Answers

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|--|---|
| 1. $4x^3 - x^2 - 7x + 2$ | $x = 3 = 19$ |
| 2. $4x^4 + 7x^2 - 12$ | 7. $x^4 - 7x^3 + 2x^2 + 6$ |
| 3. $x^3 + x^2 - 7x + 20$ | 8. $x^4 - 9x^2 + 6x - 1$ |
| 4. $-4x^3 - 11x + 18$ | 9. $2x^3 + 3x^2 + x + 15 \text{ m}^2$ |
| 5. $2x^4 - x^3 + 13x^2 - 5x + 15$ | 10. the simplified expression = $-5x + 8$, — |
| 6. the expanded product = $x^3 - 8$, the value at | |
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What is verified

11 of 12 answers machine-checked against SymPy, across 10 problems.
 — marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-APR.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 12 tagged checks.

1. Simplify $(3x^3 - 5x^2 + 2) + (x^3 + 4x^2 - 7x)$.

Solution: Adding means collecting like terms; nothing changes sign. Cubes: $3x^3 + x^3 = 4x^3$. Squares: $-5x^2 + 4x^2 = -x^2$. The $-7x$ and the $+2$ have no partner.

$4x^3 - x^2 - 7x + 2$

2. Simplify $(6x^4 + 2x^2 - 9) - (2x^4 - 5x^2 + 3)$.

Solution: Distribute the subtraction over all three terms first: $6x^4 + 2x^2 - 9 - 2x^4 + 5x^2 - 3$. Now collect: $(6 - 2)x^4$, $(2 + 5)x^2$, and $-9 - 3$.

$4x^4 + 7x^2 - 12$

3. Multiply $(x + 4)(x^2 - 3x + 5)$.

Solution: The six cells of the box are

$$\begin{array}{lll} x^3, & -3x^2, & 5x \quad (\text{top row, from } x) \\ 4x^2, & -12x, & 20 \quad (\text{bottom row, from } 4) \end{array}$$

Collecting: $-3x^2 + 4x^2 = x^2$ and $5x - 12x = -7x$.

$$\boxed{x^3 + x^2 - 7x + 20}$$

4. Simplify $2(x^3 - 4x) - 3(2x^3 + x - 6)$.

Solution: Distribute both multipliers, remembering that the -3 hits every term of the second bracket: $2x^3 - 8x - 6x^3 - 3x + 18$. Collecting, $(2 - 6)x^3$ and $(-8 - 3)x$.

$$\boxed{-4x^3 - 11x + 18}$$

5. Multiply $(2x^2 - x + 3)(x^2 + 5)$.

Solution: The box has two rows because the second factor has two terms.

$$\begin{array}{lll} 2x^4, & -x^3, & 3x^2 \quad (\text{from } x^2) \\ 10x^2, & -5x, & 15 \quad (\text{from } 5) \end{array}$$

Only the x^2 cells combine: $3x^2 + 10x^2 = 13x^2$.

$$\boxed{2x^4 - x^3 + 13x^2 - 5x + 15}$$

6. $(x - 2)(x^2 + 2x + 4)$. (a) Expand. (b) Substitute $x = 3$.

Solution (a): The cells are $x^3, 2x^2, 4x$ from x and $-2x^2, -4x, -8$ from -2 . Every middle pair cancels: $2x^2 - 2x^2 = 0$ and $4x - 4x = 0$. This is the difference-of-cubes pattern.

$$\boxed{x^3 - 8}$$

Solution (b): $3^3 - 8 = 27 - 8$. As a check, the original factors give $(3 - 2)(9 + 6 + 4) = 1 \cdot 19$, the same value — which is what substituting a number is for.

$$\boxed{19}$$

7. Simplify $(x^4 - 3x^3 + x) - (4x^3 - 2x^2 + x - 6)$.

Solution: Distribute the minus over all four terms: $x^4 - 3x^3 + x - 4x^3 + 2x^2 - x + 6$. Collecting, $-3x^3 - 4x^3 = -7x^3$, and the two x terms cancel completely, so the answer has no linear term at all.

$$\boxed{x^4 - 7x^3 + 2x^2 + 6}$$

8. Multiply $(x^2 + 3x - 1)(x^2 - 3x + 1)$.

Solution: The nine cells collect as follows. The x^3 terms ($3x^3$ and $-3x^3$) cancel; the x^2 terms are $-9x^2 - x^2 + x^2 \cdot 1 = -9x^2$; the x terms are $3x + 3x = 6x$; the constant is -1 .

$$\begin{aligned} (x^2 + 3x - 1)(x^2 - 3x + 1) &= x^4 - (3x - 1)^2 \\ &= x^4 - (9x^2 - 6x + 1) \end{aligned}$$

Seeing it as $A^2 - B^2$ with $A = x^2$ and $B = 3x - 1$ is the quick route.

$$\boxed{x^4 - 9x^2 + 6x - 1}$$

9. Garden bed $(2x + 5)$ m by $(x^2 - x + 3)$ m. Area?

Solution: Area is length times width.

$$(2x + 5)(x^2 - x + 3) = 2x^3 - 2x^2 + 6x + 5x^2 - 5x + 15$$

Collecting, $-2x^2 + 5x^2 = 3x^2$ and $6x - 5x = x$. The unit is square metres because two lengths in metres were multiplied.

$$\boxed{2x^3 + 3x^2 + x + 15 \text{ m}^2}$$

10. $(x^2 - 2x + 1)(x + 3) - (x^3 + x^2 - 5)$. (a) Simplify. (b) Explain the drop in degree.

Solution (a): Expand the product first: $(x^2 - 2x + 1)(x + 3) = x^3 + x^2 - 5x + 3$. Now subtract, distributing the minus over all three terms: $x^3 + x^2 - 5x + 3 - x^3 - x^2 + 5$. The cubes cancel and so do the squares.

$$\boxed{-5x + 8}$$

Part (b) — instructor-judged. A full-credit explanation says that the degree of a product is the *sum* of the degrees of its factors, so a degree-2 times a degree-1 gives degree 3, and that subtracting another degree-3 polynomial can cancel the leading terms and leave something of lower degree — here both the x^3 and the x^2 terms cancelled, dropping the result to degree 1. Restating the simplified expression with no comment on degree earns nothing.